

| Questions |     |      | Answers                                                                                                                                                                                                                              | Marks                                    |
|-----------|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 6         | (a) | (i)  | The electromotive force (e.m.f.) of a source is defined as the <u>amount of energy transferred from non-electrical forms of energy to electrical energy per unit charge</u> as the <u>charge passes through a complete circuit</u> . | B1                                       |
|           |     | (ii) | Potential difference<br>$= [4.5 / (4.5 + 0.5)] \times 3$<br>$= 2.7 \text{ V}$                                                                                                                                                        | A1                                       |
|           | (b) |      | Voltage across XP<br>$= (L_{xp}/L_{xy}) \times 2.7$<br>$= (0.8/1.5) \times 2.7$<br>$= 1.44 \text{ V}$<br><br>At balanced point,<br>$V_{EF} = V_{XP} = 1.44 \text{ V}$<br><br>Current $= 1.44 / 3 = 0.480 \text{ A}$                  | M1<br><br><br><br><br><br><br><br><br>A1 |
|           | (c) | (i)  | $L_{xp}$ decreases<br><br>Since $R = \rho L/A$ , when A decreases, R increases.<br>$V_{xy}$ increases so $L_{xp}$ decreases                                                                                                          | B1<br>B1                                 |
|           |     | (ii) | Since balance length decreases to potential difference $V_{xp} = V_{EF}$ , at balance length, there are no changes for the ammeter reading.                                                                                          | B1                                       |

